Physical Chemistry (I)

Lecture 13

Statistical Mechanics

In the Last Class

- Thermodynamic potentials
  - Differential forms
  - Mathematical transformations
- Thermochemistry of Gibbs energy
- Boltzmann factor

Thermodynamic Potentials

$+pV$

$-TS$

U	$H = U + pV$
$A = U - TS$	$G = U + pV - TS$

Last class

Natural Independent Variables

- Internal energy $U = U(S, V)$

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$

- Enthalpy $H = U + pV = H(S, p)$

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$$

- Helmholtz energy $A = U - TS = A(T, V)$

$$\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$$

- Gibbs energy $G = U + pV - TS = G(T, p)$

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$$

Gibbs-Helmholtz Equation

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = -\frac{H}{T^2}$$

At constant T,

$$\left(\frac{\partial(\Delta G/T)}{\partial T}\right)_p = -\frac{\Delta H}{T^2}$$

Maxwell Relations

For an exact differential $\mathrm{d}f = g\mathrm{d}x + h\mathrm{d}y$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Internal energy $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$

- Enthalpy $\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

Maxwell Relations

For an exact differential $\mathrm{d}f = g\mathrm{d}x + h\mathrm{d}y$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Helmholtz energy $\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

- Gibbs energy $\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$$

Last class

Thermodynamic Eq. of State

$$p = T\left(\frac{\partial p}{\partial T}\right)_V - \pi_T$$

Last class

Thermochemistry of G

$$G = H - TS$$

At constant T,

$$\Delta G = \Delta H - T\Delta S$$

We can define the standard Gibbs energy of reaction:

$$\Delta_{\mathrm{r}}G^{\ominus} = \Delta_{\mathrm{r}}H^{\ominus} - T\Delta_{\mathrm{r}}S^{\ominus}$$

and the standard Gibbs energy of formation:

$$\Delta_{\mathrm{f}}G^{\ominus} = \Delta_{\mathrm{f}}H^{\ominus} - T\Delta_{\mathrm{f}}S^{\ominus}$$

Last class

Pressure Dependence of G

$$\Delta G_m = \int_{p_i}^{p_f} V_m \, \mathrm{d}p$$

1. Condensed phase: $V_m \approx$ constant
2. Gas phase: $V_m = V/n = RT/p$

Last class

Boltzmann Factor

In general, for a microstate s with energy E,

$$P(s) \propto e^{-E(s)/k_\mathrm{B}T} \text{ (Boltzmann factor)}$$

Proportionality Constant

In general, for a microstate s with energy E,

$$P(s) \propto e^{-E(s)/k_\mathrm{B}T} \text{ (Boltzmann factor)}$$

or, using a constant q,

$$P(s) = \frac{1}{q} e^{-E(s)/k_\mathrm{B}T}$$

Determination of q

$$P(s) = \frac{1}{q} e^{-E(s)/k_\mathrm{B}T}$$

Using the normalization condition, $\sum_s P(s) = 1$,

$$1 = \sum_s \frac{1}{q} e^{-E(s)/k_\mathrm{B}T} = \frac{1}{q} \sum_s e^{-E(s)/k_\mathrm{B}T}$$

$$q = \sum_s e^{-E(s)/k_\mathrm{B}T}$$

(Partition function)

Brief Illustration 12A.2

Suppose that two conformations of a molecule differ in energy by 5.0 kJ mol^{-1} (corresponding to 8.3×10^{-21} J for a single molecule), so conformation A lies at energy 0 and conformation B lies at ε = 8.3×10^{-21} J. What is the proportion of molecules in conformation B at 293 K?

$E(\mathrm{A})/k_{\mathrm{B}}T$ = 0/{(1.381×10^{-23} J K^{-1})×(293 K)} = 0

$E(\mathrm{B})/k_{\mathrm{B}}T$ = (8.3×10^{-21} J)/{(1.381×10^{-23} J K^{-1})×(293 K)} = 2.05

The partition function is

$$q=\sum_{s}e^{-E(s)/k_{\mathrm{B}}T}=e^{-E(\mathrm{A})/k_{\mathrm{B}}T}+e^{-E(\mathrm{B})/k_{\mathrm{B}}T}=e^{0}+e^{-2.05}=1.13$$

Brief Illustration 12A.2

$$q = 1.13$$

Hence, the proportion of molecules in conformation B is

$$P(\mathrm{B}) = \frac{1}{q} e^{-E(\mathrm{B})/k_{\mathrm{B}}T} = \frac{1}{1.13} e^{-2.05} = 0.11$$

Inverse Temperature

$$\beta = \frac{1}{k_{\mathrm{B}}T}$$

$$q = \sum_i e^{-E_i/k_{\mathrm{B}}T} = \sum_i e^{-\beta E_i}$$

Probability Distributions

	Discrete	Continuous
Variable	$k = 1,2,\ldots,N$	x: real in $[x_1, x_2]$
Probability	$p(k)$	probability density $p(x)\mathrm{d}x$
Normalization	$\sum_{k=1}^{N} p(k) = 1$	$\int_{x_1}^{x_2} p(x)\,\mathrm{d}x = 1$
Average	$\langle k \rangle = \sum_{k=1}^{N} k p(k)$	$\langle x \rangle = \int_{x_1}^{x_2} x p(x)\,\mathrm{d}x$
Expectation	$\langle f \rangle = \sum_{k=1}^{N} f(k) p(k)$	$\langle f \rangle = \int_{x_1}^{x_2} f(x) p(x)\,\mathrm{d}x$

Average Energy

According to the probability theory, the average value of k is

$$\langle k \rangle = \sum_i p_i k_i$$

- Average energy (= internal energy)

$$\langle E \rangle = \sum_i P(E_i)E_i = \sum_i \frac{E_i e^{-\beta E_i}}{q}$$

Average Energy

$$\langle E \rangle = \sum_i \frac{E_i e^{-\beta E_i}}{q}$$

However,

$$\frac{\partial \ln q}{\partial \beta} = \frac{1}{q}\frac{\partial q}{\partial \beta} = -\frac{1}{q}\sum_i E_i e^{-\beta E_i}$$

Hence,

$$\langle E \rangle = U = -\frac{\partial \ln q}{\partial \beta}$$

What About Entropy?

$S = -k_\mathrm{B} \sum_i p_i \ln p_i$ (Shannon entropy)

Substitution of $p_i = P(E_i) = \frac{1}{q} e^{-\beta E_i}$ leads to

$$S = -k_\mathrm{B} \sum_i \frac{e^{-\beta E_i}}{q} (-\beta E_i - \ln q)$$

$$S = \beta k_\mathrm{B} \sum_i \frac{E_i e^{-\beta E_i}}{q} + \frac{k_B \ln q}{q} \sum_i e^{-\beta E_i}$$

$$S = \frac{\langle E \rangle}{T} + k_\mathrm{B} \ln q$$

Helmholtz Energy (?!)

$$S = \frac{\langle E \rangle}{T} + k_\mathrm{B} \ln q$$

Therefore,

$$-k_\mathrm{B} T \ln q = \langle E \rangle - TS = A$$

Partition Function and …

- Helmholtz energy: $A = -k_B T \ln q$
- Internal energy: $U = -\partial(\ln q)/\partial\beta$
- Entropy: $S = (U - A)/T$

"A partition function contains

all the thermodynamic information about a system."

Derived Functions

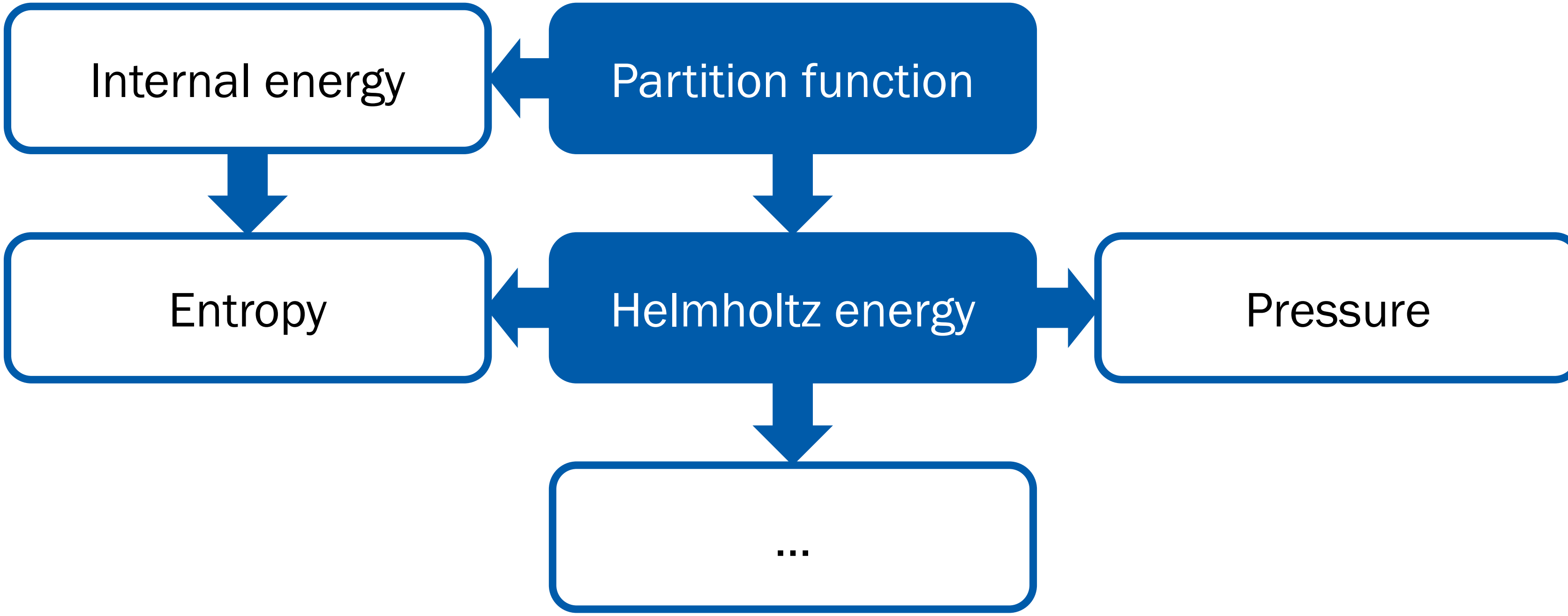
We know Helmholtz energy from the partition function,

$$A = -k_{\mathrm{B}} T \ln q$$

Using $\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$,

$$S = -\left(\frac{\partial A}{\partial T}\right)_V, \qquad p = -\left(\frac{\partial A}{\partial V}\right)_T$$

q and Thermodynamics



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Example: Spin System

Assume that we have a single spin system: up (↑) or down (↓)

$$E(\uparrow) = +\frac{\epsilon}{2}, \qquad E(\downarrow) = -\frac{\epsilon}{2}$$

The partition function is

$$q = \sum_{s} e^{-E(s)/k_{\mathrm{B}}T} = e^{-E(\uparrow)/k_{\mathrm{B}}T} + e^{-E(\downarrow)/k_{\mathrm{B}}T}$$

$$q = e^{-\epsilon/2k_{\mathrm{B}}T} + e^{+\epsilon/2k_{\mathrm{B}}T}$$

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Example: Spin System

$$q = e^{-\epsilon/2k_\mathrm{B}T} + e^{+\epsilon/2k_\mathrm{B}T}$$

- Internal energy $U = -\partial(\ln q)/\partial\beta$

$$U = -\frac{1}{q}\frac{\partial q}{\partial\beta} = -\frac{1}{q}\frac{\partial}{\partial\beta}\left(e^{-\beta\epsilon/2} + e^{\beta\epsilon/2}\right)$$

$$U = -\frac{1}{q}\left(-\frac{\epsilon}{2}e^{-\beta\epsilon/2} + \frac{\epsilon}{2}e^{\beta\epsilon/2}\right)$$

$$U = -\frac{\epsilon}{2}\frac{e^{\beta\epsilon/2} - e^{-\beta\epsilon/2}}{e^{\beta\epsilon/2} + e^{-\beta\epsilon/2}} = -\frac{\epsilon}{2}\frac{e^{\epsilon/2k_\mathrm{B}T} - e^{-\epsilon/2k_\mathrm{B}T}}{e^{\epsilon/2k_\mathrm{B}T} + e^{-\epsilon/2k_\mathrm{B}T}}$$

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Example: Spin System

$$q = e^{-\epsilon/2k_\mathrm{B}T} + e^{+\epsilon/2k_\mathrm{B}T}$$

- Helmholtz energy $A = -k_\mathrm{B}T \ln q$

$$A = -k_\mathrm{B}T \ln\left(e^{-\epsilon/2k_\mathrm{B}T} + e^{\epsilon/2k_\mathrm{B}T}\right)$$

- Entropy $S = (U - A)/T = -(\partial A/\partial T)_V$

$$S = -\frac{\epsilon}{2T}\frac{e^{\epsilon/2k_\mathrm{B}T} - e^{-\epsilon/2k_\mathrm{B}T}}{e^{\epsilon/2k_\mathrm{B}T} + e^{-\epsilon/2k_\mathrm{B}T}} + k_\mathrm{B}\ln\left(e^{-\epsilon/2k_\mathrm{B}T} + e^{\epsilon/2k_\mathrm{B}T}\right)$$

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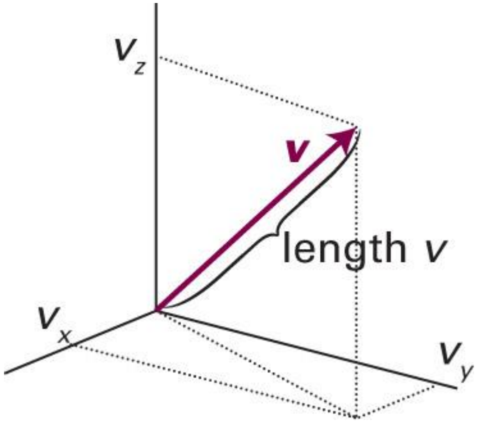
Classical Statistical Mechanics

What is a microstate of a classical particle?

Position and Velocity



Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

$$\vec{v} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \left(\frac{\mathrm{d}x}{\mathrm{d}t}, \frac{\mathrm{d}y}{\mathrm{d}t}, \frac{\mathrm{d}z}{\mathrm{d}t}\right)$$

With the velocity, we can define:

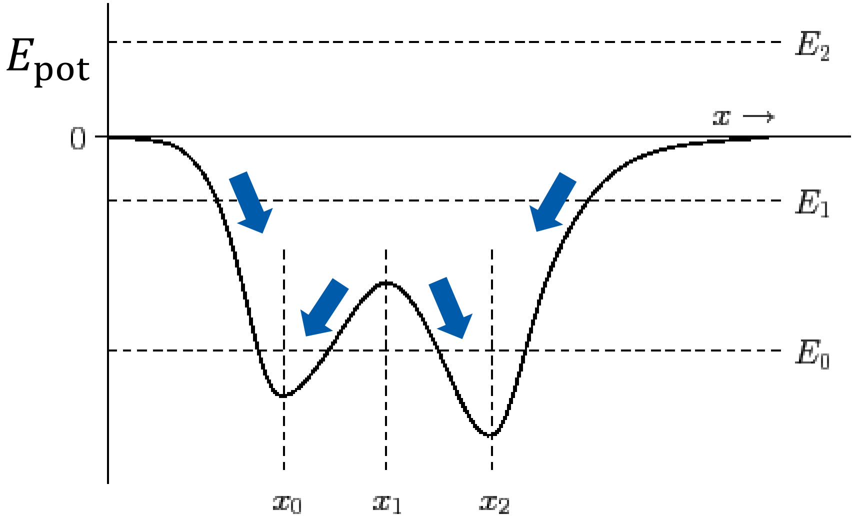
- Speed $v = |\vec{v}| = \left(v_x^2 + v_y^2 + v_z^2\right)^{1/2}$
- Kinetic energy $E_\mathrm{k} = mv^2/2 = m\left(v_x^2 + v_y^2 + v_z^2\right)/2$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., the chemist's toolkit 3, Sketch 1)

Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Force $\vec{F}$: negative slope of potential energy $(-\nabla E_{\text{pot}})$



(Image: http://farside.ph.utexas.edu/teaching/336k/Newtonhtml/node16.html)

Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Acceleration $\vec{a}$: second time-derivative of position

$$\vec{a} = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\vec{r}}{\mathrm{d}t^2}$$

$$-\nabla E_{\mathrm{pot}} = m\frac{\mathrm{d}^2\vec{r}}{\mathrm{d}t^2}$$

given

Second-Order ODE

$$\frac{\mathrm{d}^2\vec{r}}{\mathrm{d}t^2} = -\frac{\nabla E_{\mathrm{pot}}}{m}$$

Two conditions are required to obtain the full dynamics:

(initial) position and (initial) velocity (= momentum)

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Phase Space

Space of all possible microstates of a system

- Classical mechanics:

Space of all possible values of

position and momentum variables

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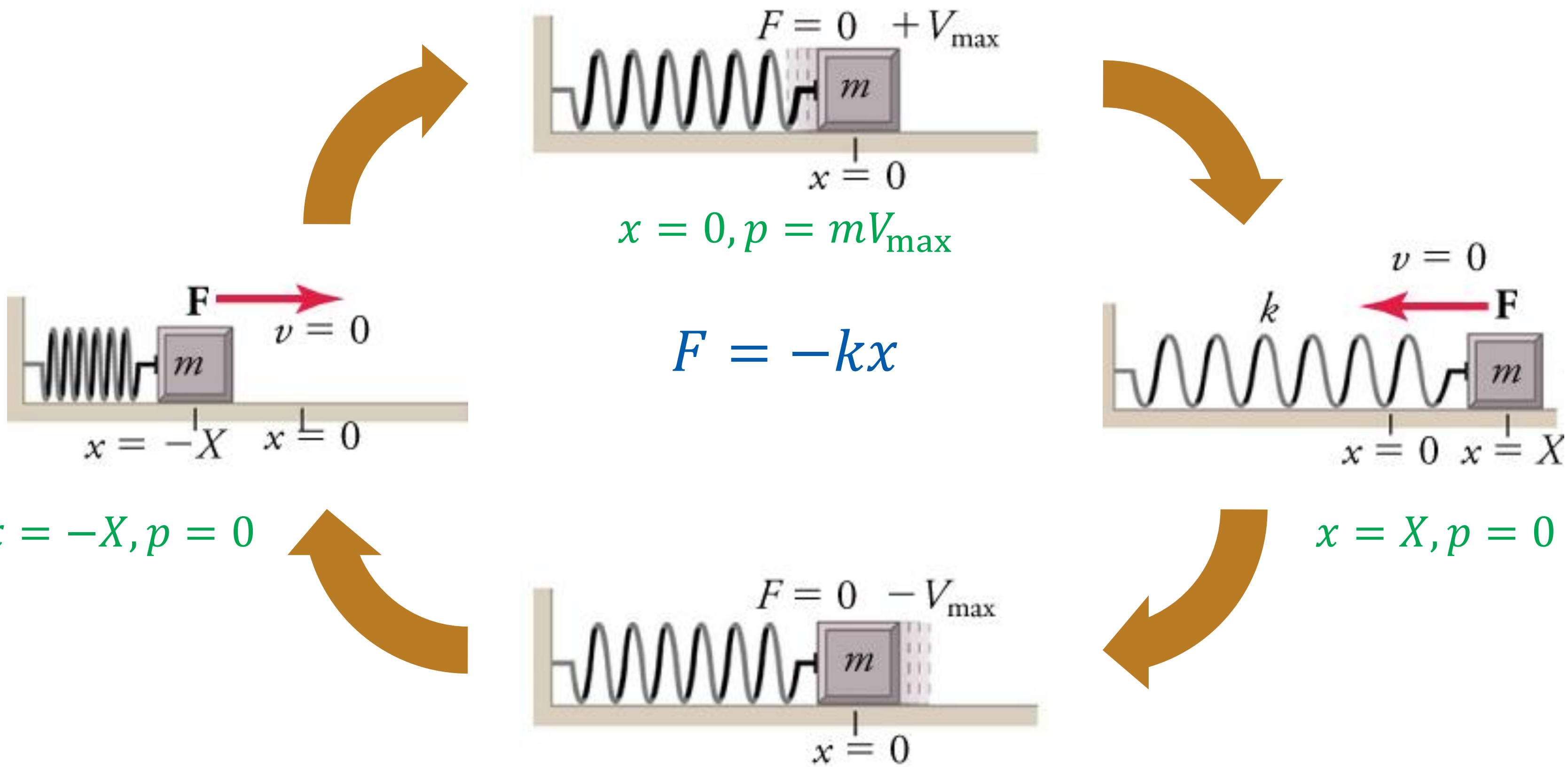
1-Dimensional Spring



$x = 0, p = mV_{\text{max}}$

$x = -X, p = 0$

$$F = -kx$$

$x = X, p = 0$

$x = 0, p = -mV_{\text{max}}$

(Image: https://openstax.org/books/physics/pages/5-5-simple-harmonic-motion)

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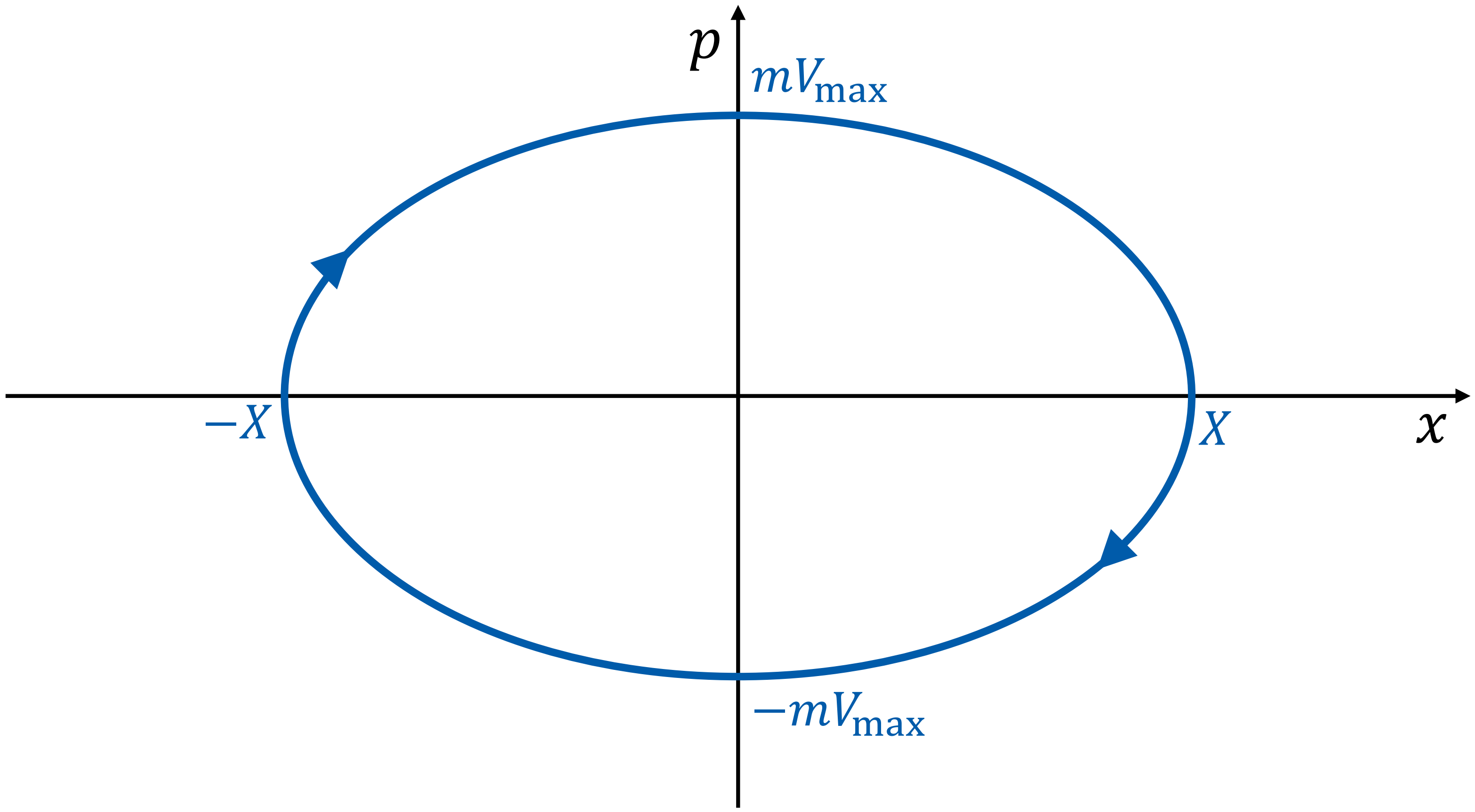
Phase Space Representation



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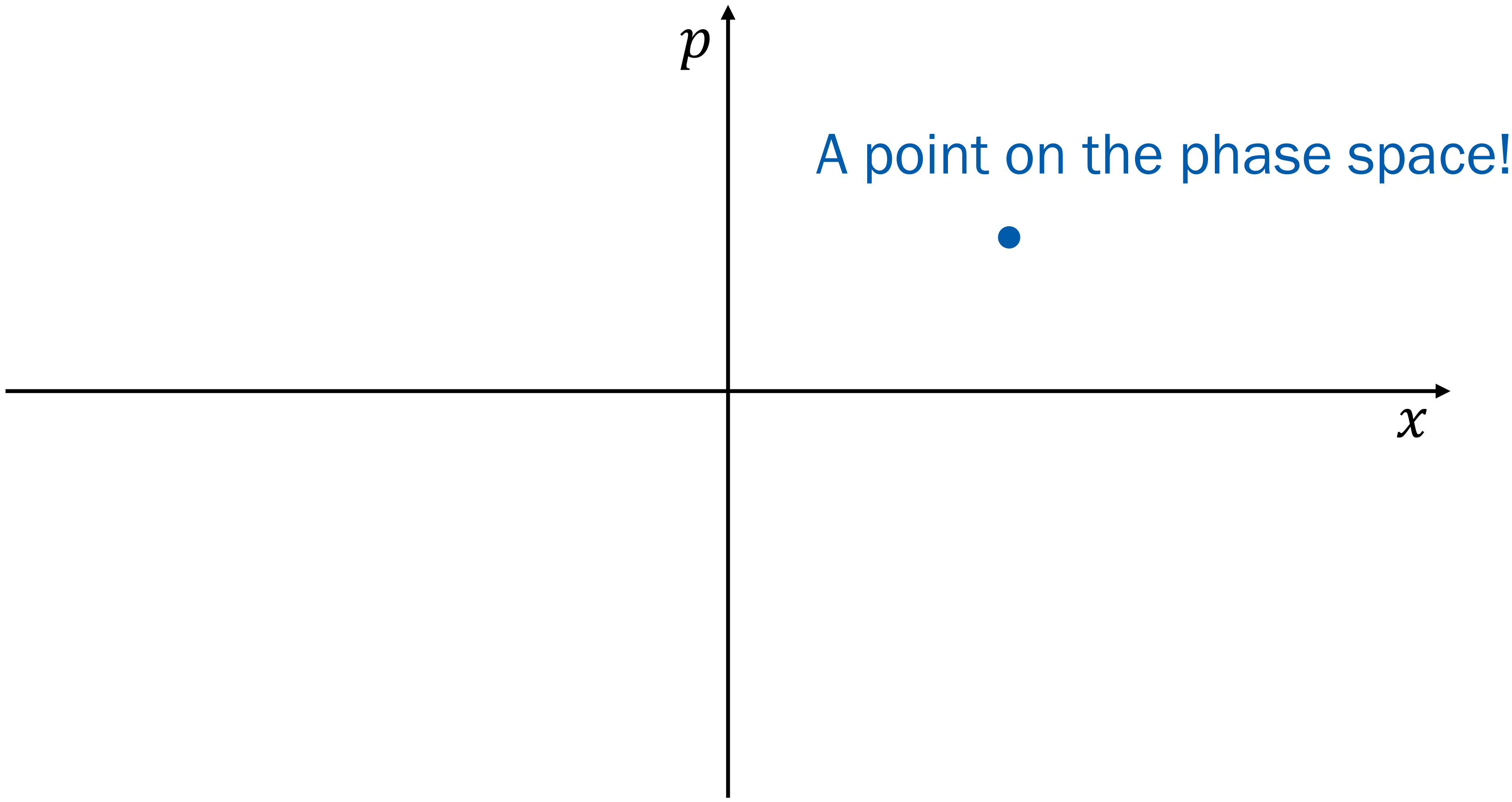
Microstate of Classical Particle



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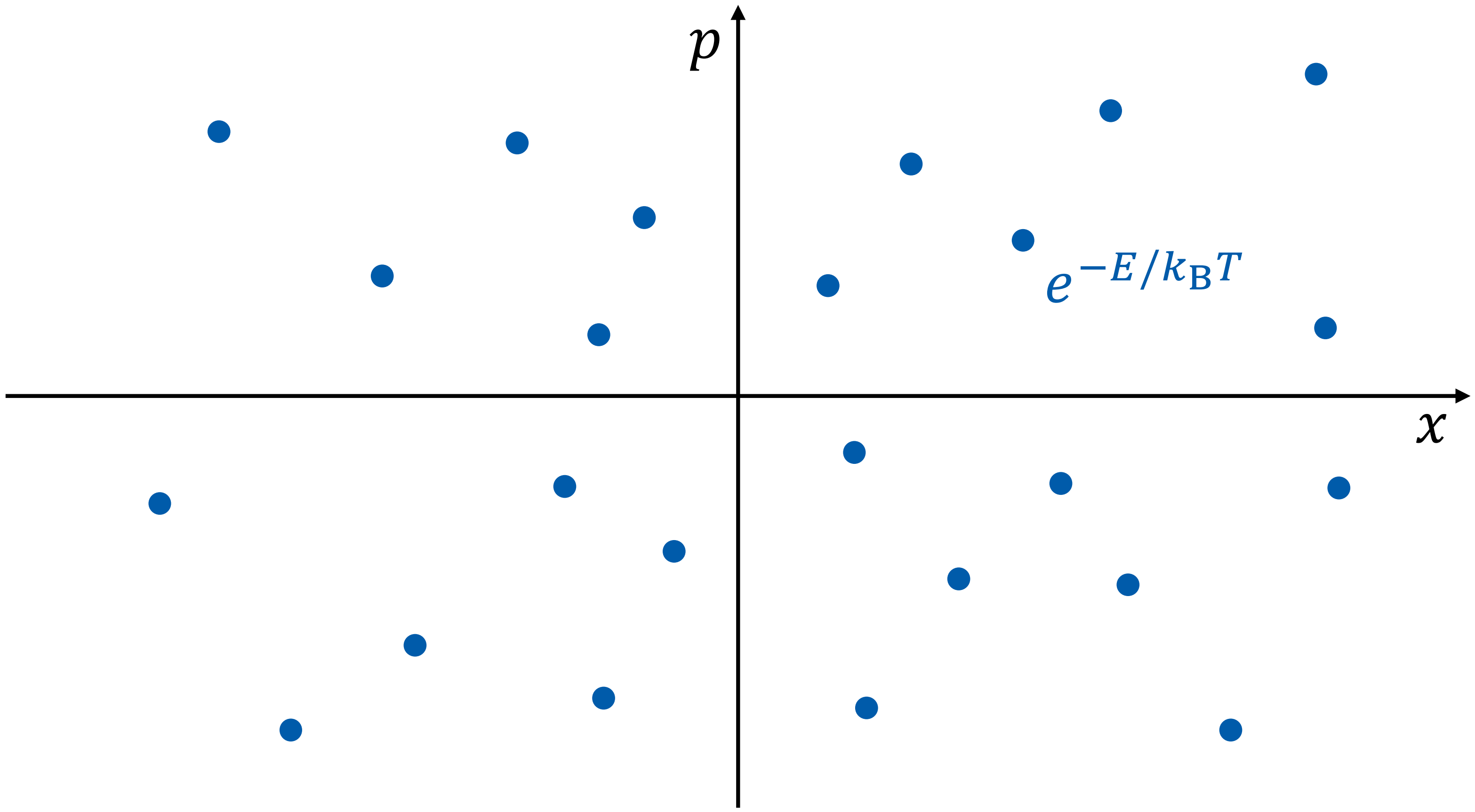
Partition Function



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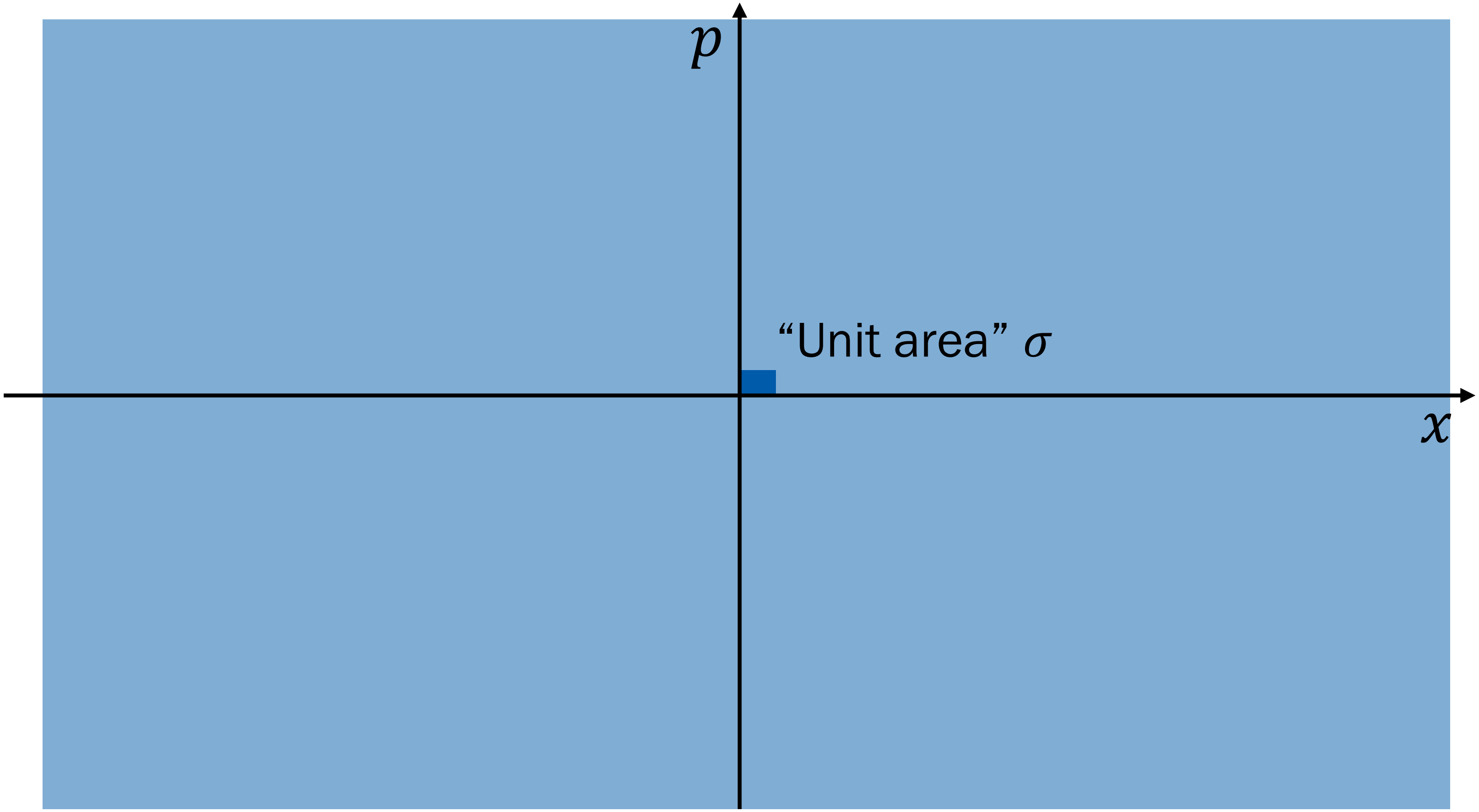
Partition Function



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Partition Function

$$q = \sum_{s} e^{-E(s)/k_{\mathrm{B}}T} \quad \rightarrow \quad q = \frac{1}{\sigma} \int \int e^{-E(x,p)/k_{\mathrm{B}}T} \, \mathrm{d}x \mathrm{d}p$$

for a 1-dimensional single particle

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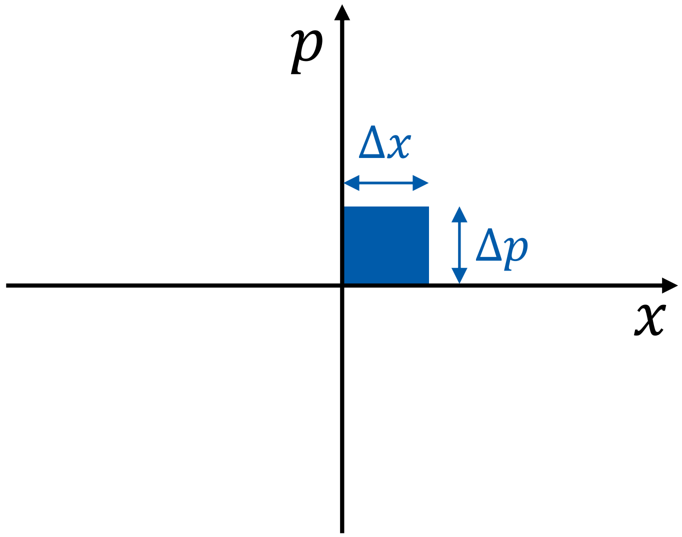
What is σ?

From Heisenberg's uncertainty principle,

$\Delta x \Delta p \geq h$ (Planck's constant)



Take $\sigma = h$!

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Partition Function

$$q = \frac{1}{h}\int\int e^{-E(x,p)/k_\mathrm{B}T}\,\mathrm{d}x\mathrm{d}p$$

- Example: 1-dimensional spring

$$E(x,p) = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kx^2 + \frac{p^2}{2m}$$

$$q = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{1}{k_\mathrm{B}T}\left(\frac{1}{2}kx^2+\frac{p^2}{2m}\right)}\,\mathrm{d}x\mathrm{d}p$$

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1-Dimensional Spring

$$q = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{1}{k_{\mathrm{B}}T}\left(\frac{1}{2}kx^2+\frac{p^2}{2m}\right)}\,\mathrm{d}x\mathrm{d}p$$

$$q = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}-\frac{p^2}{2mk_{\mathrm{B}}T}}\,\mathrm{d}x\mathrm{d}p$$

$$q = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}} \times e^{-\frac{p^2}{2mk_{\mathrm{B}}T}}\,\mathrm{d}x\mathrm{d}p$$

$$q = \frac{1}{h}\left[\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}}\,\mathrm{d}x\right]\left[\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mk_{\mathrm{B}}T}}\,\mathrm{d}p\right]$$

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Gaussian Integrals

$$q = \frac{1}{h}\left[\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_\mathrm{B}T}}\,\mathrm{d}x\right]\left[\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mk_\mathrm{B}T}}\,\mathrm{d}p\right]$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2}\,\mathrm{d}x = \sqrt{\pi/\alpha}$,

$$q = \frac{1}{h}\sqrt{\frac{2\pi k_\mathrm{B}T}{k}}\sqrt{2m\pi k_\mathrm{B}T} = \frac{2\pi k_\mathrm{B}T}{h}\sqrt{\frac{m}{k}}$$

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Into Higher Dimensions

- Single 1-dimensional particle

$$q = \frac{1}{h} \int \int e^{-E(x,p)/k_\mathrm{B}T} \, \mathrm{d}x\mathrm{d}p$$

- Single 3-dimensional particle

$$q = \frac{1}{h^3} \iiint \iiint e^{-E(x,y,z,p_x,p_y,p_z)/k_\mathrm{B}T} \, \mathrm{d}x\mathrm{d}y\mathrm{d}z\mathrm{d}p_x\mathrm{d}p_y\mathrm{d}p_z$$

$$\rightarrow \; q = \frac{1}{h^3} \int \int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T} \, \mathrm{d}\vec{r}\mathrm{d}\vec{p}$$

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Higher and Higher

- Single 3-dimensional particle

$$q = \frac{1}{h^3} \int \int e^{-E(\vec{r},\vec{p})/k_{\mathrm{B}}T} \, \mathrm{d}\vec{r}\mathrm{d}\vec{p}$$

- N 3-dimensional (independent) particles

$$Q = q_1 \times q_2 \times q_3 \times q_4 \times \cdots \times q_N$$

$$Q = \frac{1}{h^{3N}} \left[\int \int e^{-E(\vec{r},\vec{p})/k_{\mathrm{B}}T} \, \mathrm{d}\vec{r}\mathrm{d}\vec{p} \right]^N$$

Indistinguishability

- N 3-dimensional distinguishable particles:

$$Q = \frac{1}{h^{3N}}\left[\int\int e^{-E(\vec{r},\vec{p})/k_{\mathrm{B}}T}\,\mathrm{d}\vec{r}\mathrm{d}\vec{p}\right]^{N}$$

- N 3-dimensional indistinguishable particles:

$$Q = \frac{1}{N!\,h^{3N}}\left[\int\int e^{-E(\vec{r},\vec{p})/k_{\mathrm{B}}T}\,\mathrm{d}\vec{r}\mathrm{d}\vec{p}\right]^{N}$$

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Calculation of Q

For indistinguishable particles,

$$Q = \frac{1}{N!\,h^{3N}}\left[\int\int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\,\mathrm{d}\vec{r}\mathrm{d}\vec{p}\right]^N = \frac{1}{N!}q^N$$

where

$$q = \frac{1}{h^3}\int\int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\,\mathrm{d}\vec{r}\mathrm{d}\vec{p}$$